The Missing Chapter: Environmental Impact Belongs in Social Implications of Computing

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CS 492, Social Implications of Computing, examined bias, discrimination, privacy, and regulation with rigour and depth. Yet a critical dimension remained largely unaddressed: the environmental impact of technology. This essay argues that this omission is not merely a gap in coverage but a structural limitation, because the social and environmental consequences of computing are inseparable. The ICT sector contributes an estimated 3.8% of global greenhouse gas emissions, a figure projected to rise sharply [1]. Global e-waste reached 53.6 million metric tonnes in 2019 alone [4]. Cryptocurrency mining produces carbon emissions on the scale of mid-sized nations [2]. And the supply chains sustaining the hardware underlying all of these systems are implicated in environmental destruction and human rights abuses in mining regions of the developing world [3]. By weaving environmental considerations into the existing curriculum, particularly through the lens of environmental justice and ICT life-cycle assessment, the course would better equip students to understand and challenge the full scope of technology's social consequences.

Keywords: environmental impact, ICT emissions, e-waste, cryptocurrency, supply chains, environmental justice, right to repair, life-cycle assessment, social implications

I. INTRODUCTION

CS 492's exploration of the social implications of computing has been an enlightening journey, dissecting the ethical dilemmas woven into the fabric of our technological world. The course grappled with the specters of bias, discrimination, privacy breaches, and the ever-evolving landscape of regulation. Yet, within this intricate landscape, a critical thread remains largely unraveled: the environmental impact of technology. Integrating this dimension into the curriculum would be paramount to fostering a truly holistic understanding of technology's role in our society.

The environmental and the social are not separate domains. They are entangled in ways that make it impossible to fully understand algorithmic bias, global data collection, or technology regulation without also understanding the physical sub-

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strate that makes computing possible, and the ecological and human costs of maintaining it. This essay identifies the environmental impact of technology as the most significant missed topic in CS 492, describes what research reveals about that impact, and explains how it connects to every major theme the course did address.

II. THE CARBON FOOTPRINT OF ICT

Beyond the social concerns the course meticulously examined, the environmental footprint of technology casts a long shadow. The vast energy consumption and carbon emissions associated with the ICT sector are a cause for alarm. A landmark study estimates the ICT industry's contribution to global greenhouse gas emissions at 3.8%, projected to rise to as high as 14% of the global total by 2040 if current trends continue [1]. These figures demand not only a critical shift towards renewable energy sources, but also stricter regulations on data center efficiency. Incorporating LCA into the design phase of technology products could drastically reduce their environmental footprint by encouraging the development of energy-efficient devices.

The sheer scale of data center infrastructure underlying the digital services that course discussions took for granted, from cloud computing to social media platforms to the AI systems studied in the context of healthcare and criminal justice, means that no discussion of these technologies is complete without acknowledging the energy systems that power them.

III. ENVIRONMENTAL JUSTICE AND ALGORITHMIC BIAS

The conversation on bias in AI algorithms, a topic examined in the context of healthcare and criminal justice, takes on a new dimension when considering its environmental implications. These algorithms can perpetuate environmental injustices. Consider a scenario where clean energy solutions are prioritized for wealthier areas based on socioeconomic data, leaving marginalized communities further burdened by pollution. Examining the environmental impact assessments of AI projects could be a powerful tool in mitigating such biases and ensuring a more equitable distribution of environmental resources and burdens.

Environmental justice, the principle that no community should bear a disproportionate share of environmental harm, connects directly to the equity frameworks the course applied to algorithmic discrimination. The same structural inequalities that produce biased healthcare algorithms also determine which communities live adjacent to data centers, which regions host the server farms that power global digital services, and which populations absorb the pollution from inadequate e-waste processing.

IV. SUPPLY CHAINS, RARE EARTH EXTRACTION, AND HUMAN RIGHTS

Delving into the global supply chains that fuel the tech industry reveals an uncomfortable connection between everyday technology consumption and distant environmental and human rights abuses. The extraction of rare earth metals, vital components in electronic devices, often comes at a devastating cost for the environment and local communities, particularly in developing nations. A 2021 report by the European Environmental Bureau documents the severe water and air pollution caused by mines extracting lithium, cobalt, and nickel, metals essential to both consumer electronics and the electric vehicles often promoted as environmental solutions [3]. These environmental concerns are intricately linked to the human rights abuses surrounding child labour and unsafe working conditions often found in these mining operations.

This expands the conversation on global injustice beyond algorithmic bias, drawing a direct connection between our seemingly innocuous technology habits and the exploitation of people and resources in developing nations. The data colonialism discussion the course touched upon has a physical, material counterpart: the extraction of minerals from those same communities whose data is collected without adequate consent or benefit.

V. THE E-WASTE CRISIS

The issue of e-waste also demands attention. Designed for planned obsolescence rather than longevity, electronic devices accumulate in electronic graveyards, generating a global crisis. A record 53.6 million metric tonnes of e-waste were generated worldwide in 2019 alone, up 21% in just five years, with the majority disposed of through improper methods like open burning that releases toxic substances including lead, mercury, and cadmium into the environment [4]. Integrating discussions on right-to-repair legislation and fostering business models that incentivize product reuse could significantly reduce e-waste generation and pave the way for a more sustainable approach to technology consumption.

The right-to-repair movement connects directly to the course's discussions of corporate power and consumer rights. The same regulatory frameworks examined in relation to privacy law and algorithmic accountability are also the mechanisms through which governments might require manufacturers to design repairable, longer-lived devices.

VI. CRYPTOCURRENCY AND THE HIDDEN COSTS OF DIGITAL SERVICES

The environmental impact extends beyond hardware. The very essence of digital services contributes to environmental degradation. Cryptocurrency mining is particularly notorious for its energy demands. A 2023 study published in *Earth's Future* by United Nations scientists found that Bitcoin mining relied on fossil energy sources for the majority of its electricity supply, with coal alone accounting for 45% of the energy mix during the 2020–2021 period, and that the carbon emissions from Bitcoin mining in China alone in 2021–2022 would require planting approximately two billion trees to offset [2]. Furthermore, the constant data transfers and cloud storage associated with our reliance on online services contribute to the overall environmental burden of the ICT industry.

This connects to the course's treatment of surveillance and data privacy: the same data-intensive systems that raise civil liberties concerns also carry substantial environmental costs that fall disproportionately on communities near fossil fuel infrastructure.

VII. EMERGING TECHNOLOGIES: OPPORTUNITIES AND TRADE-OFFS

Looking beyond immediate concerns, the course would benefit from examining the potential long-term environmental effects of emerging technologies. AI presents both opportunities and challenges for environmental sustainability. On one hand, AI could be used to develop more efficient renewable energy solutions and optimize resource management. Conversely, the immense computational power required for advanced AI research could exacerbate the energy demands of the ICT industry. Critically evaluating these trade-offs would encourage students to resist both uncritical technophilia and reflexive rejection of emerging tools.

A similar analysis applies to electric vehicles. Their promotion as a straightforward environmental solution obscures the fact that the mining of lithium and cobalt for their batteries raises the same environmental and human rights concerns as the mining of rare earth metals for consumer electronics [3]. By examining the full environmental life cycle of these technologies, students develop a more sophisticated understanding of what sustainability actually requires.

VIII. INTEGRATION WITH THE EXISTING CURRICULUM

The environmental dimension should not be a standalone addition but should permeate existing discussions. The course's examination of facial recognition technology could be expanded to consider the environmental footprint of the vast data centers required to store and process facial recognition data. The discussion of algorithmic bias in online advertising could be reframed to explore how targeted advertising drives excessive consumption with both social and environmental consequences.

This integrated approach reflects the actual structure of the problems. Environmental degradation and social injustice are not parallel crises but deeply entangled ones. The communities most

affected by e-waste processing, rare earth mining, and proximity to data center infrastructure are overwhelmingly the same communities already marginalized by algorithmic bias and inadequate data privacy protection [3, 4].

IX. CONCLUSION

The urgency of the climate crisis necessitates integrating the environmental impact of technology into CS 492. By studying these consequences alongside the social concerns the course explored, students gain a more nuanced understanding of technology's role in society. This broadened perspective empowers students to advocate for solutions that promote sustainability alongside equity and social justice in the digital age. It also fosters a sense of individual responsibility, encouraging conscious choices about technology consumption and support for companies that prioritize environmental responsibility in their practices. Ultimately, the aim is not just to understand the complex tapestry of social implications, but to become responsible co-creators of a future where technology serves as a tool for progress rather than a source of environmental degradation.

ACRONYMS

AI: Artificial Intelligence: a branch of computer science concerned with building systems that perform tasks typically requiring human intelligence; referenced in the context of both its potential to optimize energy systems and its substantial computational energy demands 2, 3, 4

e-waste: Electronic waste: discarded electrical or electronic devices including computers, smartphones, and household appliances; a record 53.6 million metric tonnes were generated globally in 2019, with only 17.4% formally recycled 1, 2, 3, 4

ICT: Information and Communication Technology: the broad sector encompassing computing hardware, software, telecommunications, internet infrastructure, and data centers; estimated to contribute 3.8% of global greenhouse gas emissions as of 2018, with projections suggesting this could rise to 14% by 2040 1, 2, 3, 4

LCA: Life-Cycle Assessment: a systematic methodology for evaluating the environmental impacts associated with all stages of a product's life, from raw material extraction through manufacturing, use, and end-of-life disposal; advocated as a tool for reducing the environmental footprint of technology products at the design phase 2, 4

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